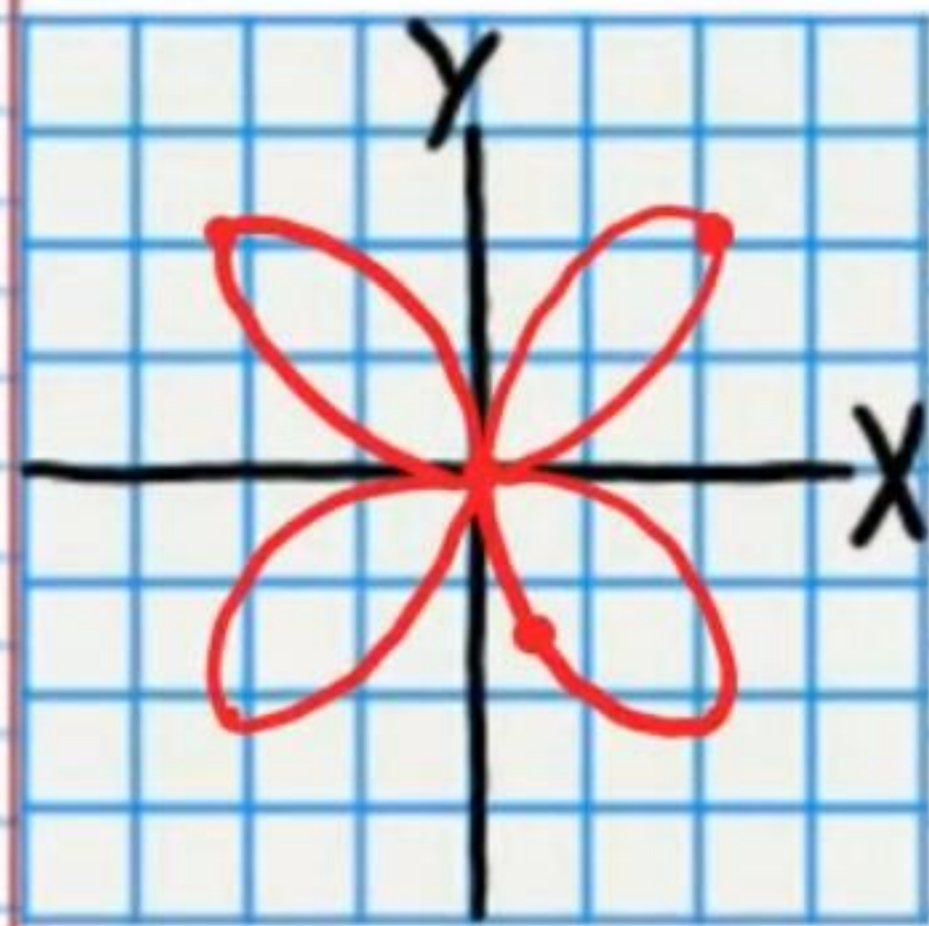


Graphing Polar Equations and Converting to Rectangular Form

Graph the following equations. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

① $r = 3\sin(2\theta)$



1st Point

$\theta = 0^\circ$

$r = 3\sin(2(0^\circ))$

$r = 3\sin 0^\circ$

$r = 3(0)$

$r = 0$

$(0, 0^\circ)$

2nd Point

$\theta = 45^\circ$

$r = 3\sin(2(45^\circ))$

$r = 3\sin 90^\circ$

$r = 3(1)$

$r = 3$

$(3, 45^\circ)$

3rd Point

$\theta = -45^\circ$

$r = 3\sin(2(-45^\circ))$

$r = 3\sin(-90^\circ)$

$r = 3(-1)$

$r = -3$

$(-3, -45^\circ)$

4th Point

$\theta = 105^\circ$

$r = 3\sin(2(105^\circ))$

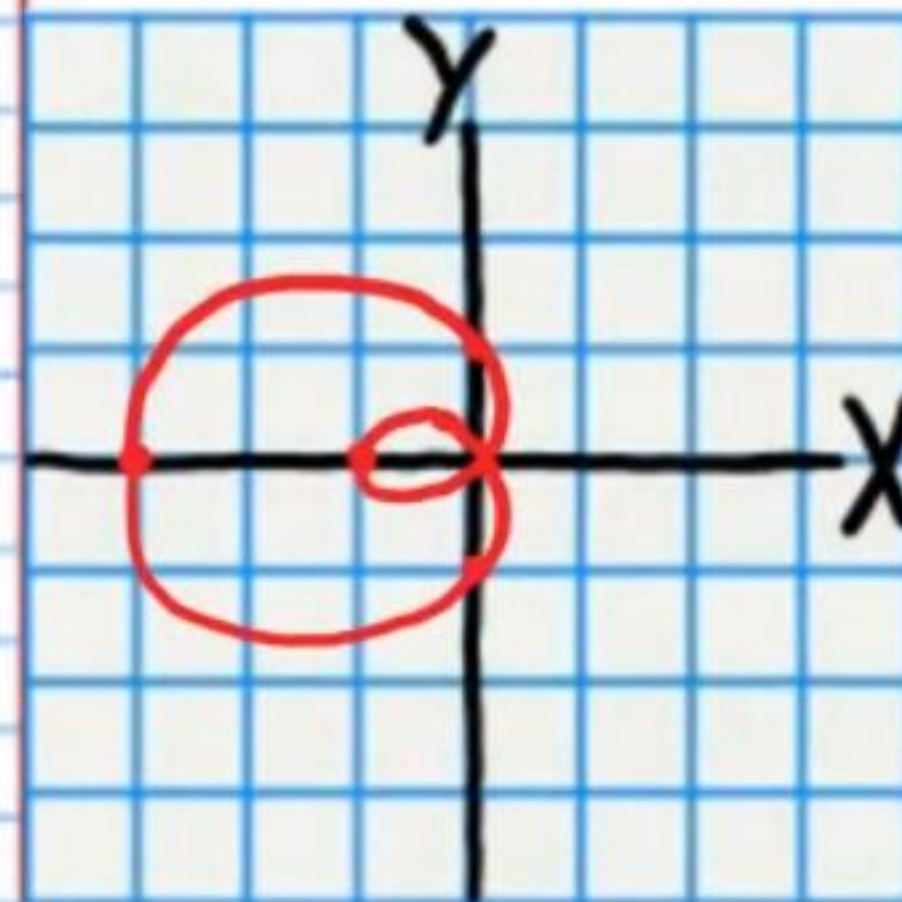
$r = 3\sin 210^\circ$

$r = 3(-\frac{1}{2})$

$r = -\frac{3}{2}$

$(-\frac{3}{2}, 105^\circ)$

② $r = 1 - 2\cos\theta$



1st Point

$\theta = 0^\circ$

$r = 1 - 2\cos 0^\circ$

$r = 1 - 2(1)$

$r = 1 - 2$

$r = -1$

$(-1, 0^\circ)$

2nd Point

$\theta = 90^\circ$

$r = 1 - 2\cos 90^\circ$

$r = 1 - 2(0)$

$r = 1 - 0$

$r = 1$

$(1, 90^\circ)$

3rd Point

$\theta = 180^\circ$

$r = 1 - 2\cos 180^\circ$

$r = 1 - 2(-1)$

$r = 1 + 2$

$r = 3$

$(3, 180^\circ)$

4th Point

$\theta = -90^\circ$

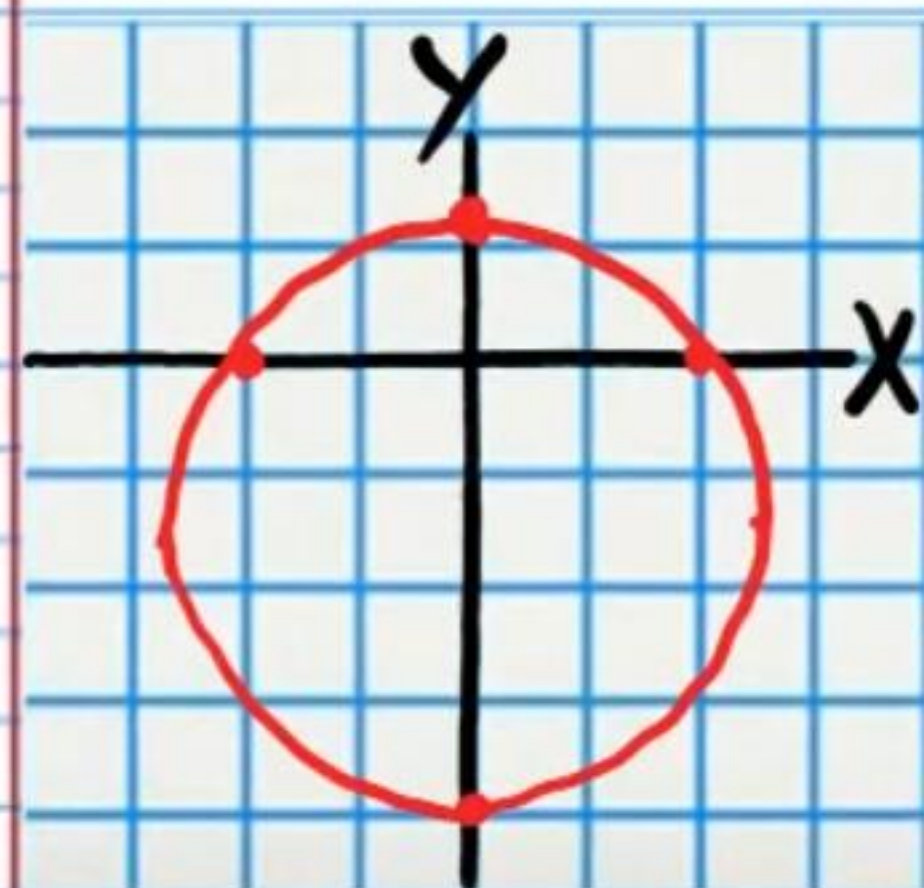
$r = 1 - 2\cos(-90^\circ)$

$r = 1 - 2(0)$

$r = 1$

$(1, -90^\circ)$

③ $r = \frac{4}{2 + \sin \theta}$



1st Point

$$\theta = 0^\circ$$

$$r = \frac{4}{2 + \sin 0^\circ}$$

$$r = \frac{4}{2 + 0}$$

$$r = 2$$

$$(2, 0^\circ)$$

2nd Point

$$\theta = 90^\circ$$

$$r = \frac{4}{2 + \sin 90^\circ}$$

$$r = \frac{4}{2 + 1}$$

$$r = \frac{4}{3}$$

$$\left(\frac{4}{3}, 90^\circ\right)$$

3rd Point

$$\theta = -90^\circ$$

$$r = \frac{4}{2 + \sin(-90^\circ)}$$

$$r = \frac{4}{2 + (-1)}$$

$$r = 4$$

$$(4, -90^\circ)$$

4th Point

$$\theta = 180^\circ$$

$$r = \frac{4}{2 + \sin 180^\circ}$$

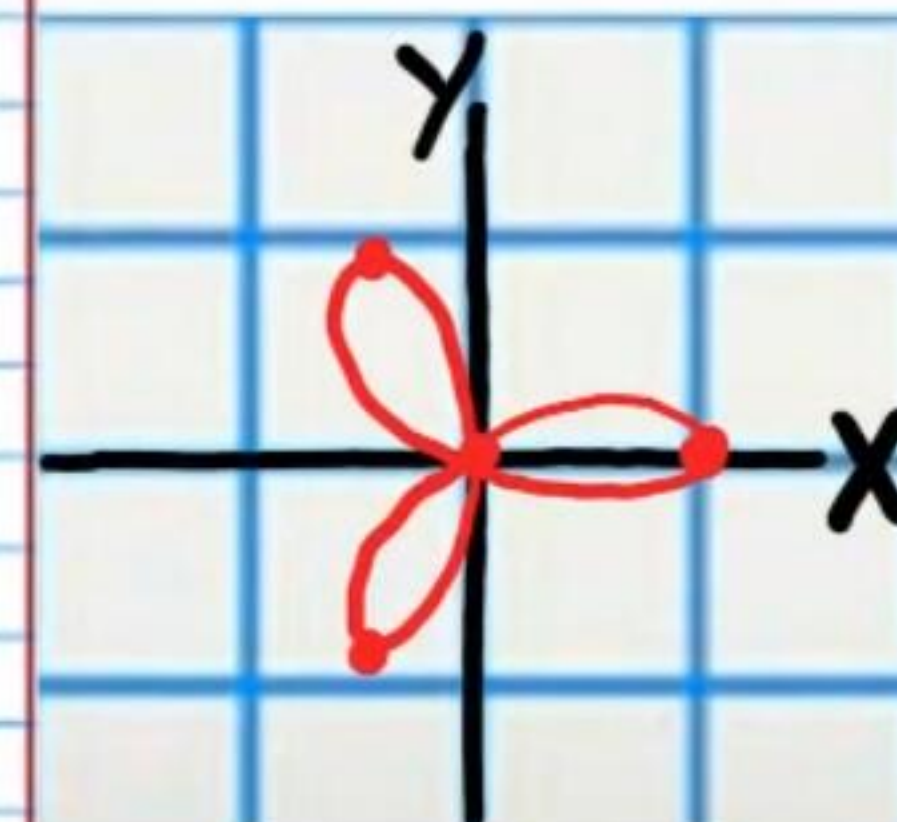
$$r = \frac{4}{2 + 0}$$

$$r = 2$$

$$(2, 180^\circ)$$

★ Change the window in your calculator.

④ $r = \cos(3\theta)$



1st Point

$$\theta = 0^\circ$$

$$r = \cos(3(0^\circ))$$

$$r = \cos 0^\circ$$

$$r = 1$$

$$(1, 0^\circ)$$

2nd Point

$$\theta = 90^\circ$$

$$r = \cos(3(90^\circ))$$

$$r = \cos 270^\circ$$

$$r = 0$$

$$(0, 90^\circ)$$

3rd Point

$$\theta = 60^\circ$$

$$r = \cos(3(60^\circ))$$

$$r = \cos 180^\circ$$

$$r = -1$$

$$(-1, 60^\circ)$$

4th Point

$$\theta = -60^\circ$$

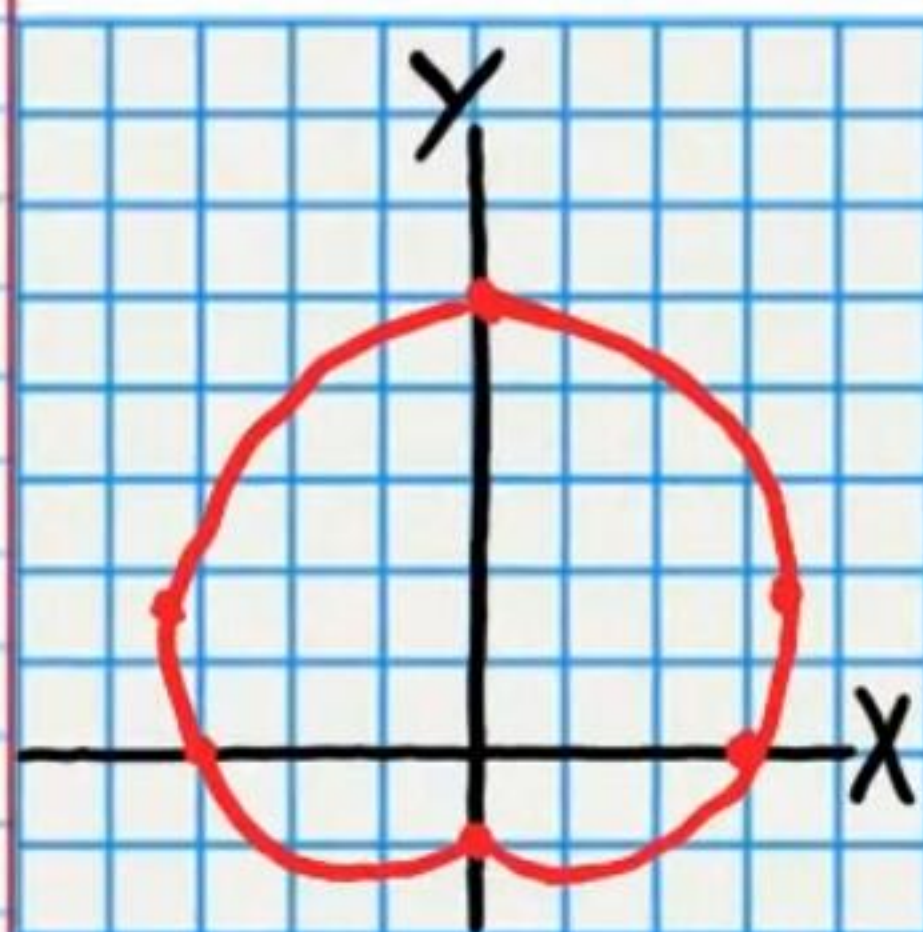
$$r = \cos(3(-60^\circ))$$

$$r = \cos(-180^\circ)$$

$$r = -1$$

$$(-1, -60^\circ)$$

⑤ $r = 3 + 2\sin\theta$



1st Point

$\theta = 0^\circ$

$r = 3 + 2\sin 0^\circ$

$r = 3 + 2(0)$

$r = 3 + 0$

$r = 3$

$(3, 0^\circ)$

2nd Point

$\theta = 90^\circ$

$r = 3 + 2\sin 90^\circ$

$r = 3 + 2(1)$

$r = 3 + 2$

$r = 5$

$(5, 90^\circ)$

3rd Point

$\theta = -90^\circ$

$r = 3 + 2\sin(-90^\circ)$

$r = 3 + 2(-1)$

$r = 3 - 2$

$r = 1$

$(1, -90^\circ)$

4th Point

$\theta = 30^\circ$

$r = 3 + 2\sin 30^\circ$

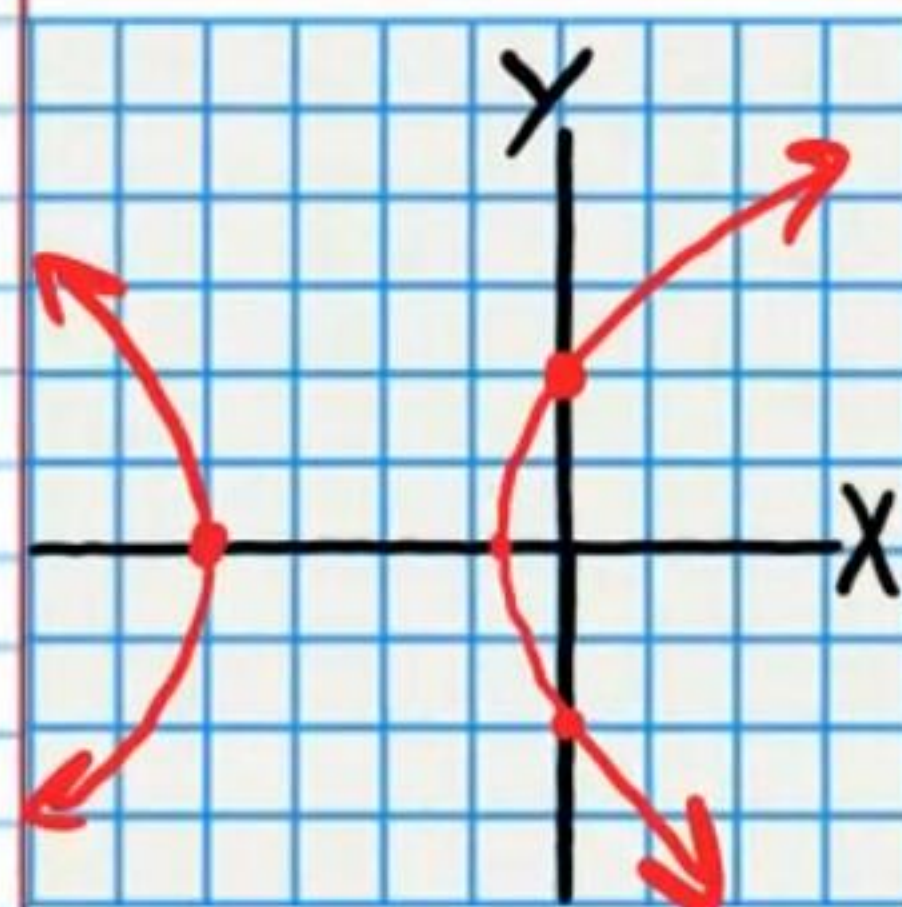
$r = 3 + 2\left(\frac{1}{2}\right)$

$r = 3 + 1$

$r = 4$

$(4, 30^\circ)$

⑥ $r = \frac{4}{2 - 3\cos\theta}$



1st Point

$\theta = 0^\circ$

$r = \frac{4}{2 - 3\cos 0^\circ}$

$r = \frac{4}{2 - 3(1)}$

$r = -4$

$(-4, 0^\circ)$

2nd Point

$\theta = 90^\circ$

$r = \frac{4}{2 - 3\cos 90^\circ}$

$r = \frac{4}{2 - 3(0)}$

$r = 2$

$(2, 90^\circ)$

3rd Point

$\theta = -90^\circ$

$r = \frac{4}{2 - 3\cos(-90^\circ)}$

$r = \frac{4}{2 - 3(0)}$

$r = 2$

$(2, -90^\circ)$

4th Point

$\theta = 180^\circ$

$r = \frac{4}{2 - 3\cos 180^\circ}$

$r = \frac{4}{2 - 3(-1)}$

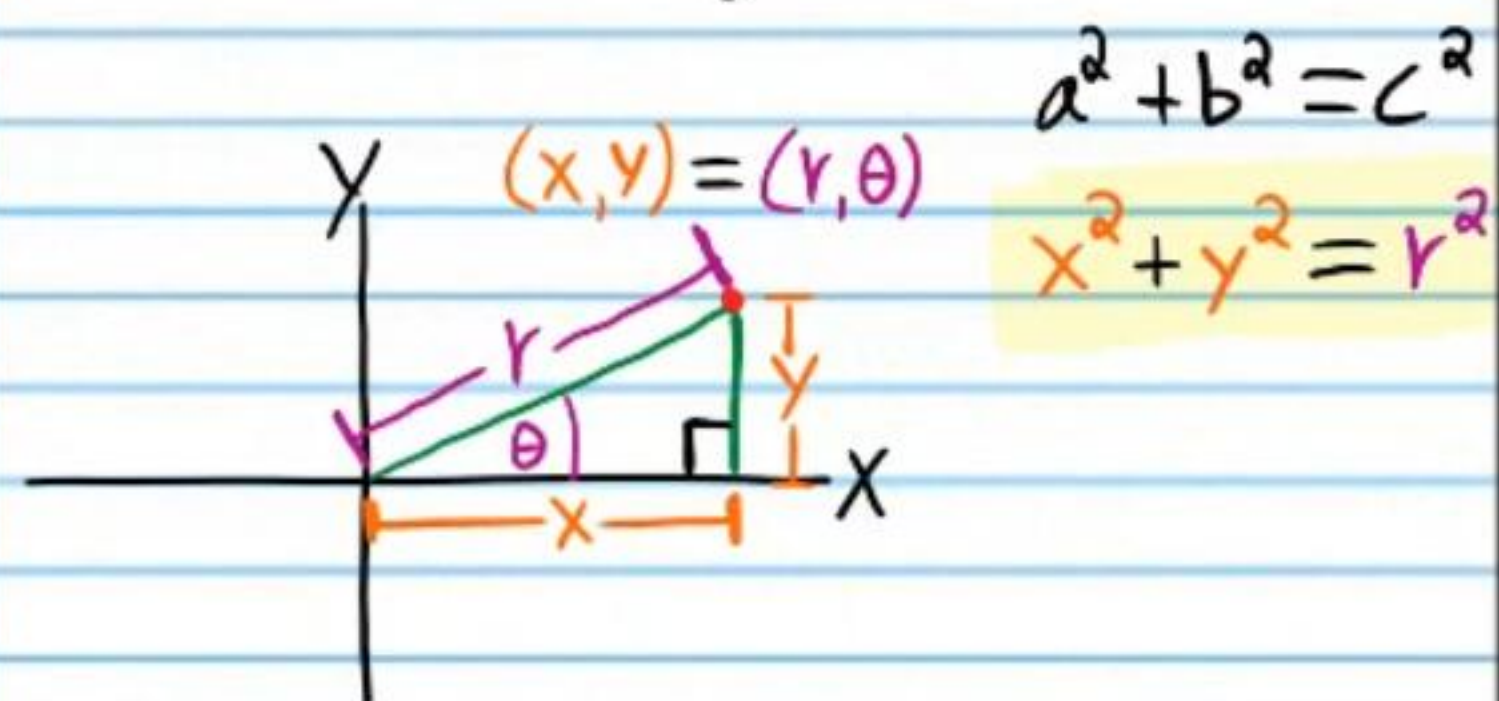
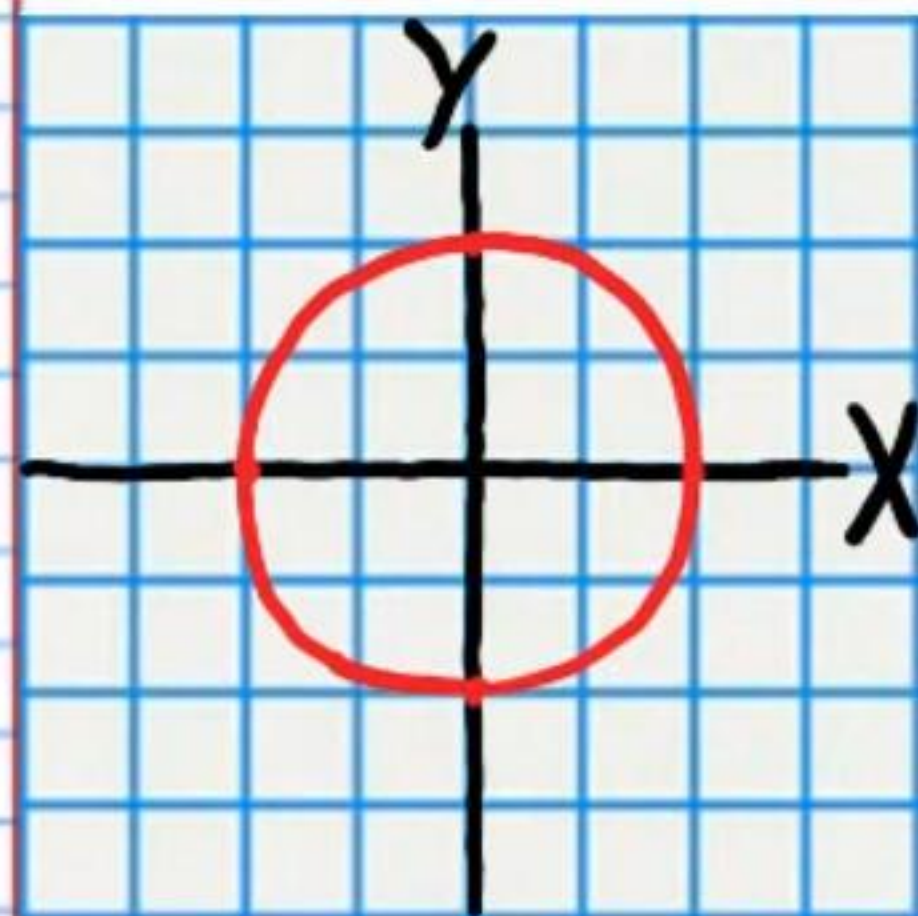
$r = \frac{4}{5}$

$\left(\frac{4}{5}, 180^\circ\right)$

• ⑦ $r = 2$

Convert to rectangular form.

Use the Pythagorean Theorem



$$r = 2$$

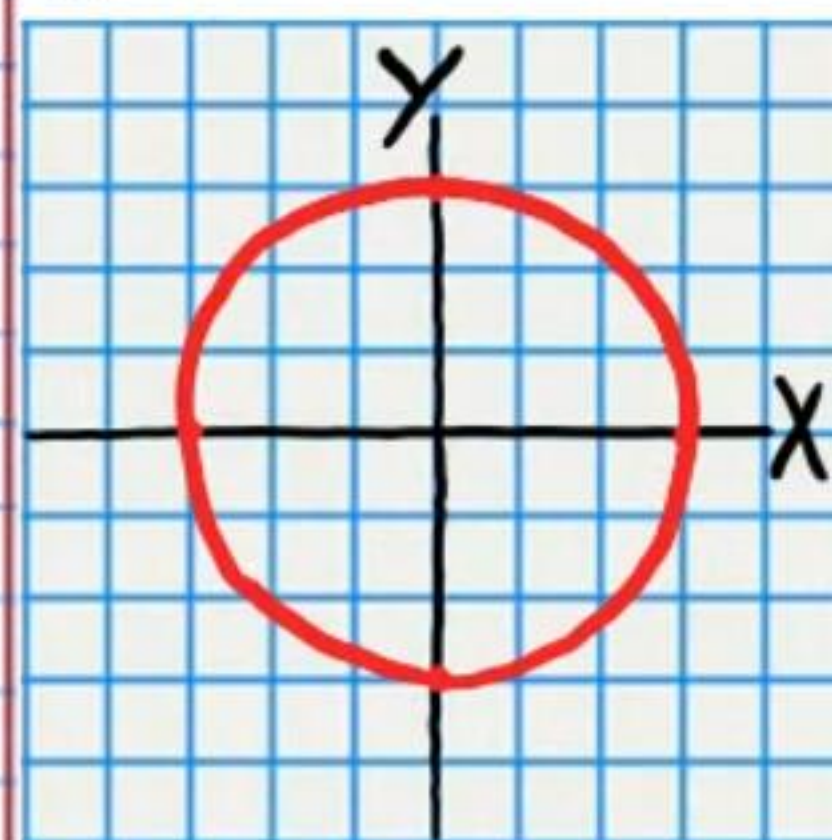
$$r^2 = 2^2$$

$$r^2 = 4$$

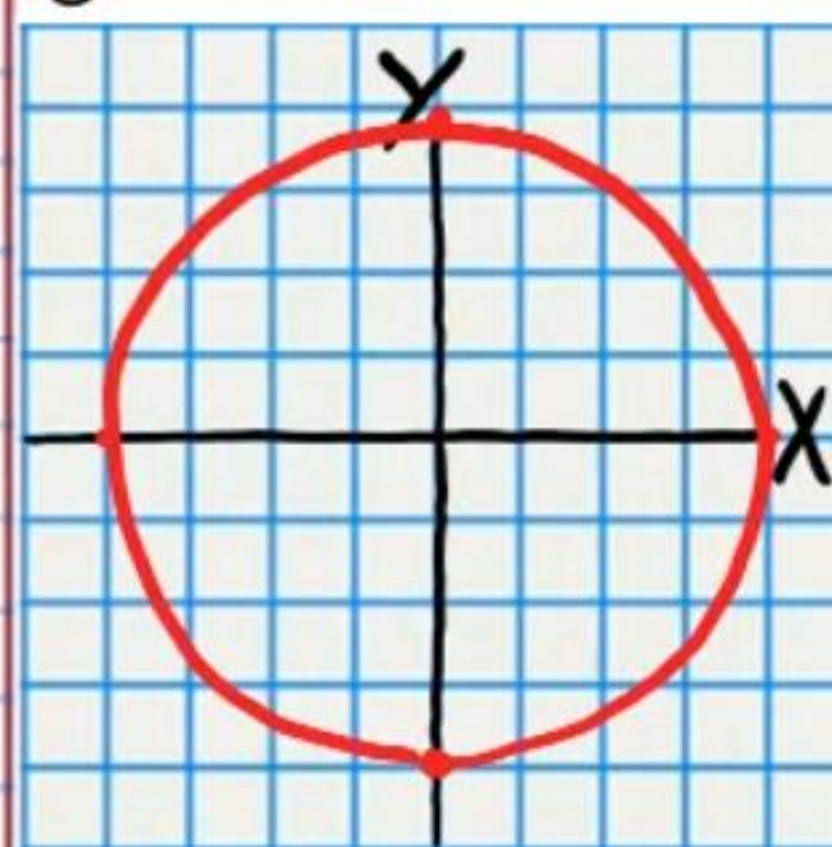
$$x^2 + y^2 = 4$$

$$(x-0)^2 + (y-0)^2 = 2^2$$

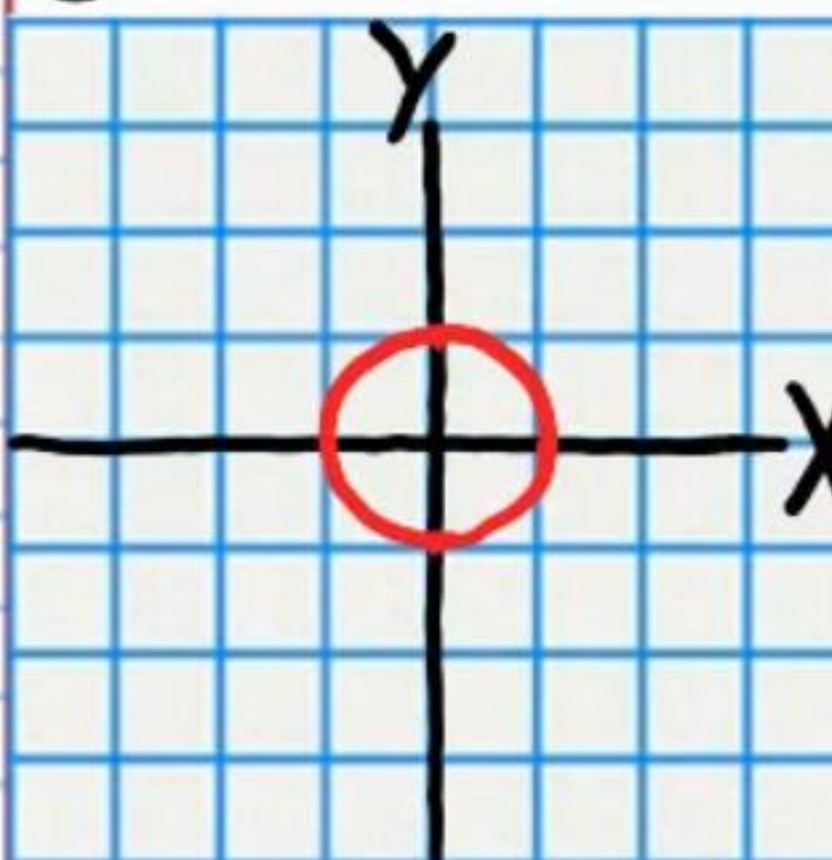
• ⑧ $r = 3$



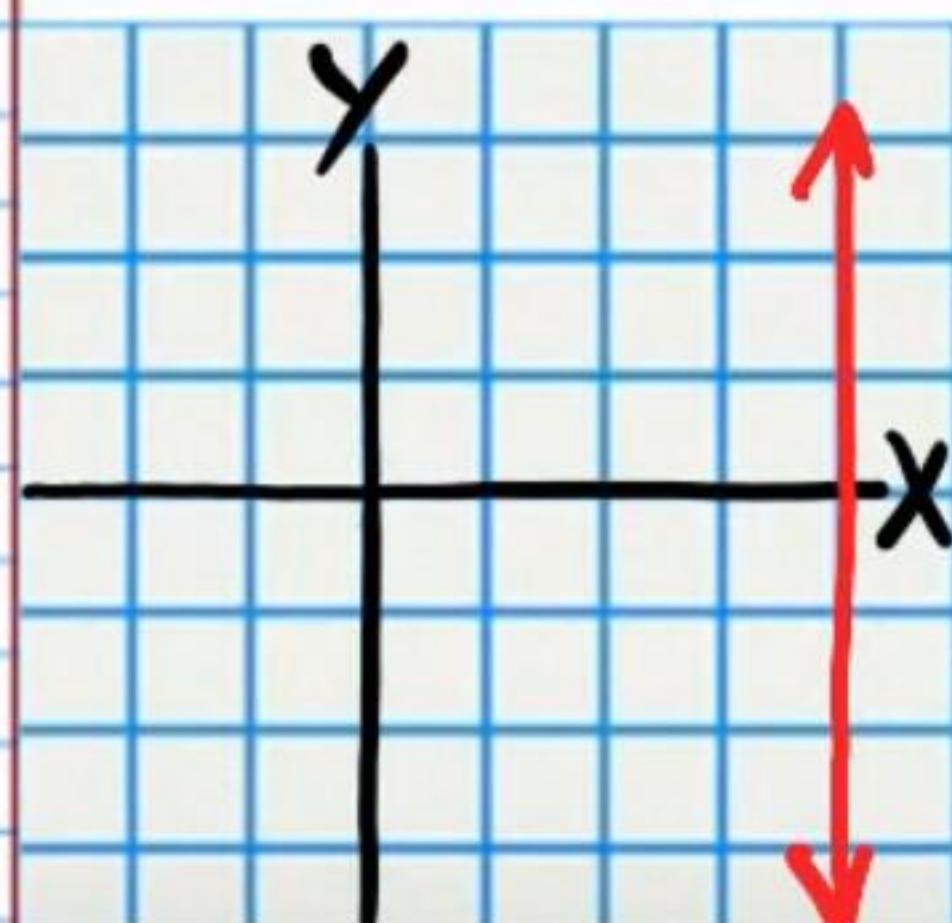
• ⑨ $r = 4$



⑩ $r = 1$

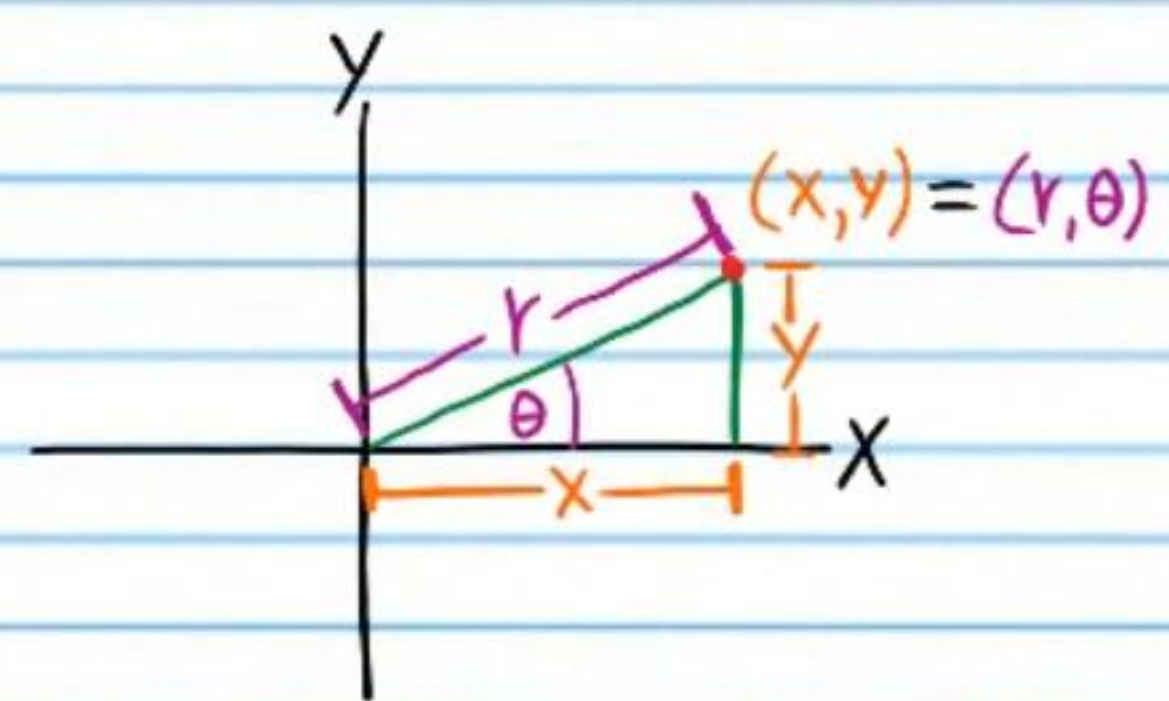


• ⑪ $\cos \theta = \frac{4}{r}$



Convert to rectangular form.

Use Trigonometry



$$\cos \theta = \frac{x}{r} \quad \sin \theta = \frac{y}{r}$$

$$x = r \cos \theta \quad y = r \sin \theta$$

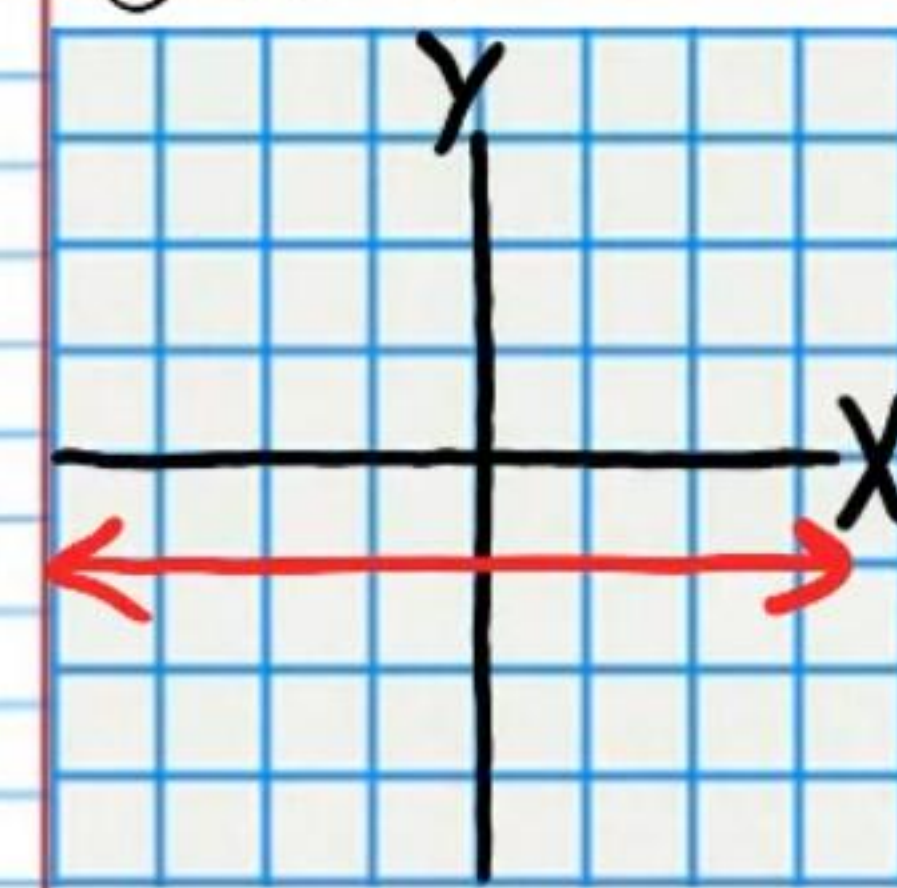
$$\cos \theta = \frac{4}{r}$$

$$r \cdot \cos \theta = \frac{4}{r} \cdot r$$

$$r \cos \theta = 4$$

$$x = 4$$

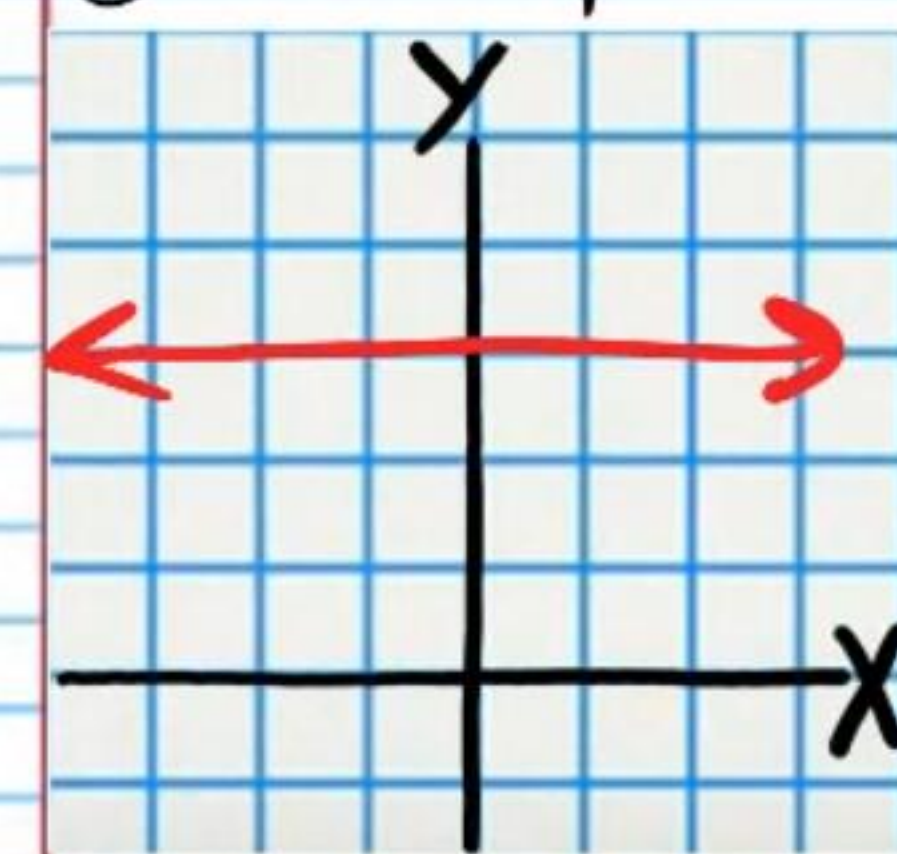
⑫ $r \sin \theta = -1$



$$r \sin \theta = -1$$

$$y = -1$$

• ⑬ $\sin \theta = \frac{3}{r}$

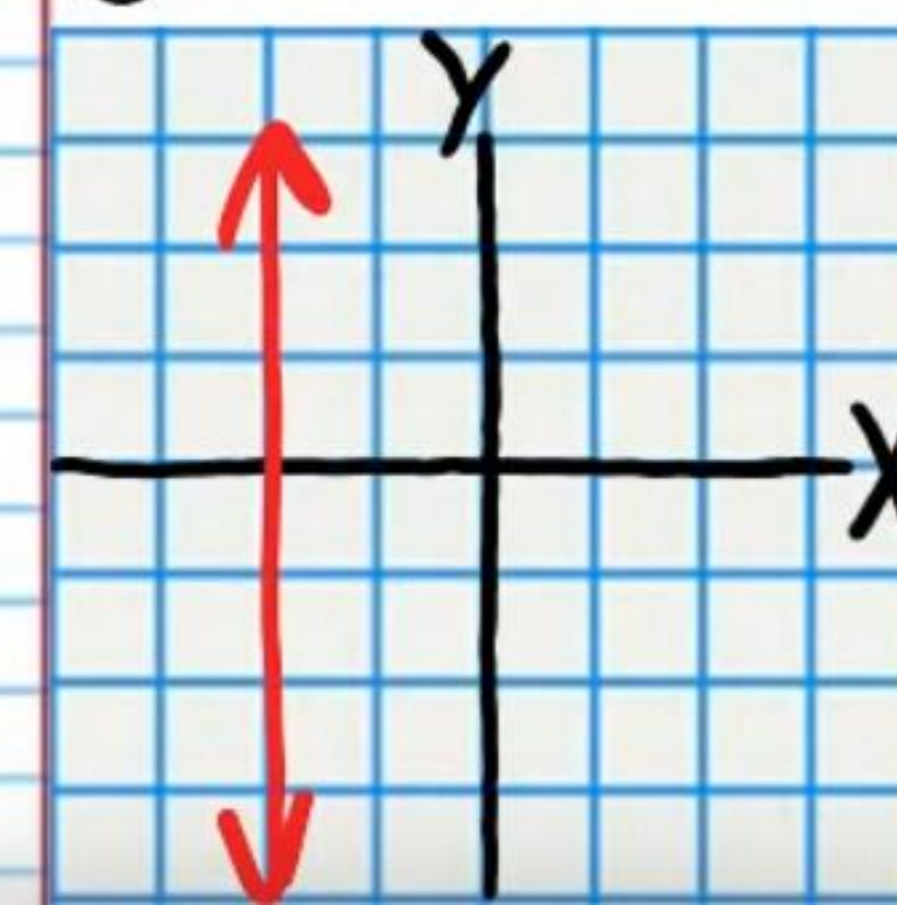


$$\sin \theta = \frac{3}{r}$$

$$r \sin \theta = 3$$

$$y = 3$$

⑭ $r \cos \theta = -2$

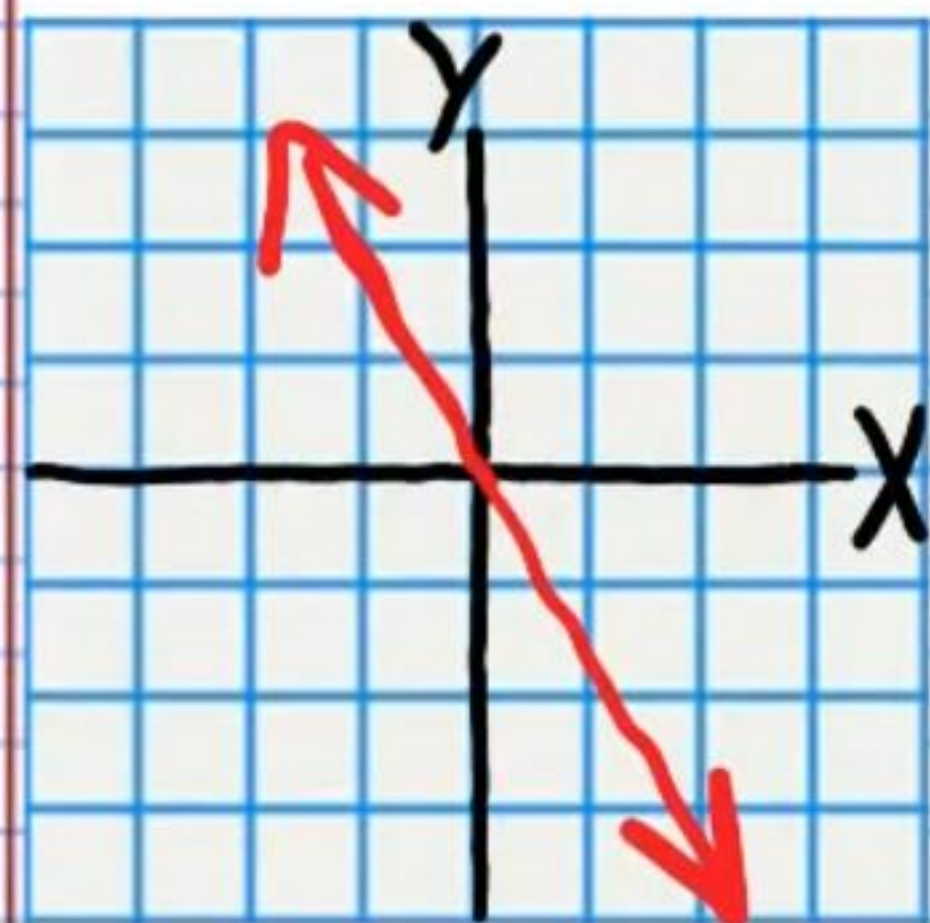


$$r \cos \theta = -2$$

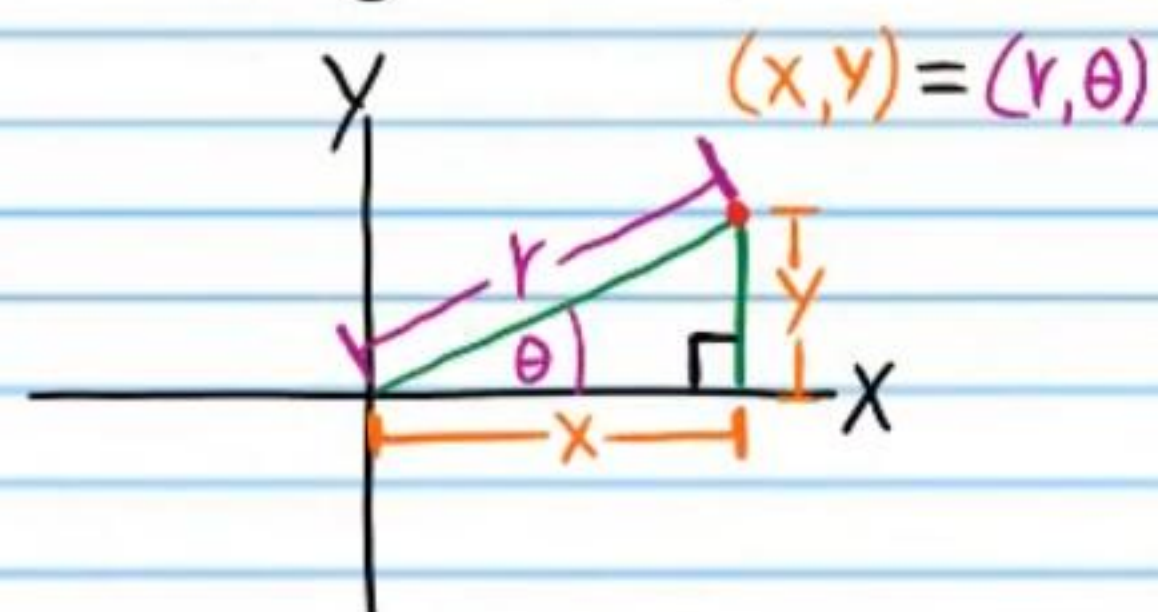
$$x = -2$$

• ⑮ $\theta = 120^\circ$

Convert to rectangular form.



Use trigonometry

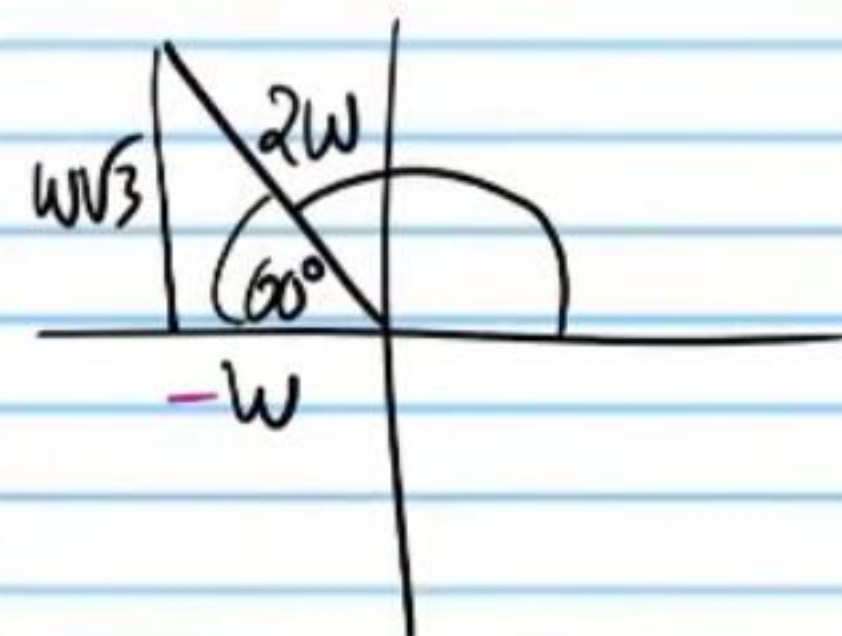


$$\tan \theta = \frac{y}{x}$$

$$\tan 120^\circ = \frac{y}{x}$$

$$-\sqrt{3} = \frac{y}{x}$$

$$-x\sqrt{3} = y$$



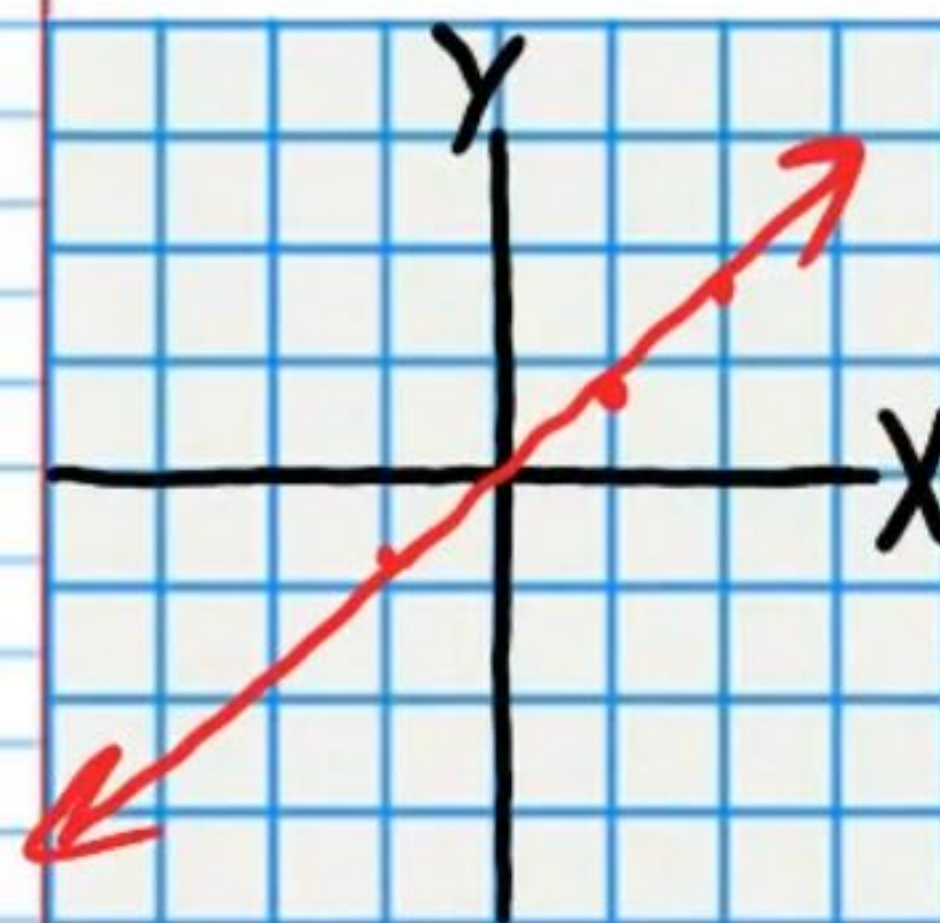
⑯ $\theta = 220^\circ$

$$\tan \theta = \frac{y}{x}$$

$$\tan 220^\circ = \frac{y}{x}$$

$$0.839 \approx \frac{y}{x}$$

$$0.839x \approx y$$



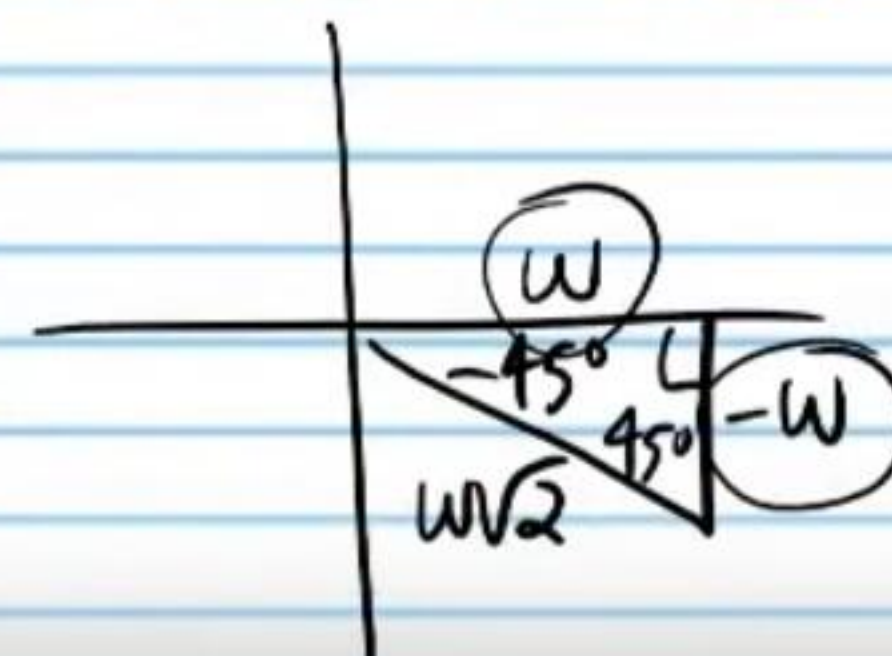
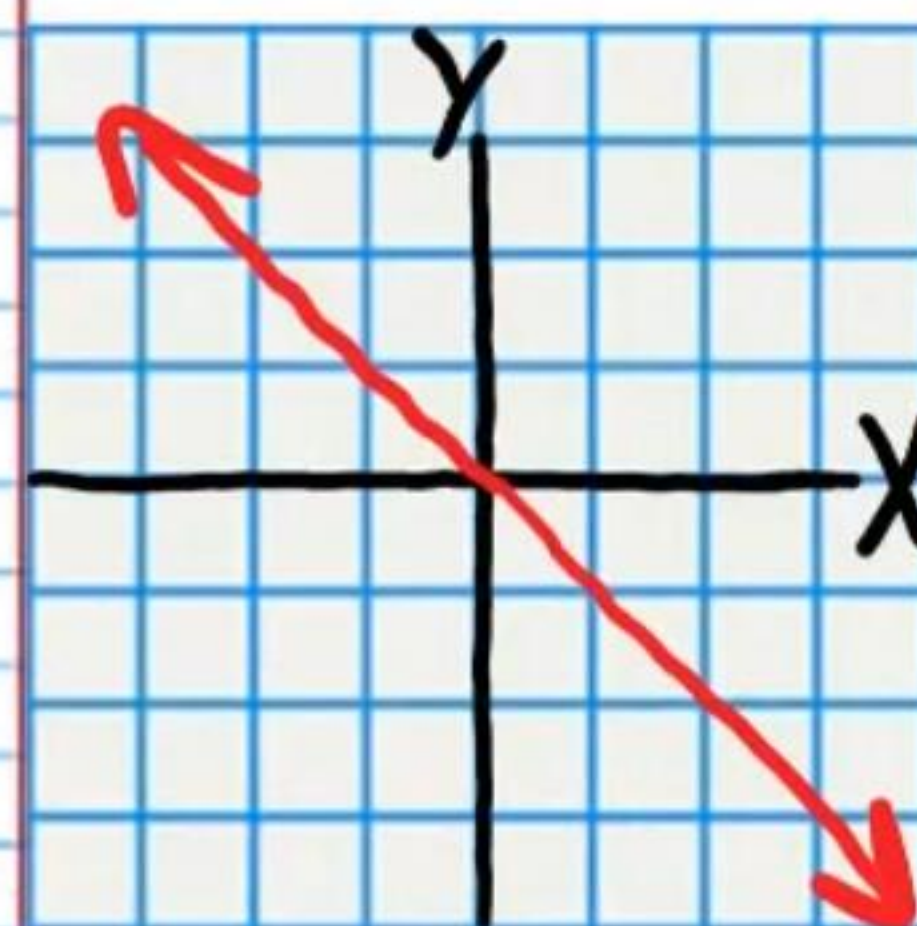
⑰ $\theta = -45^\circ$

$$\tan \theta = \frac{y}{x}$$

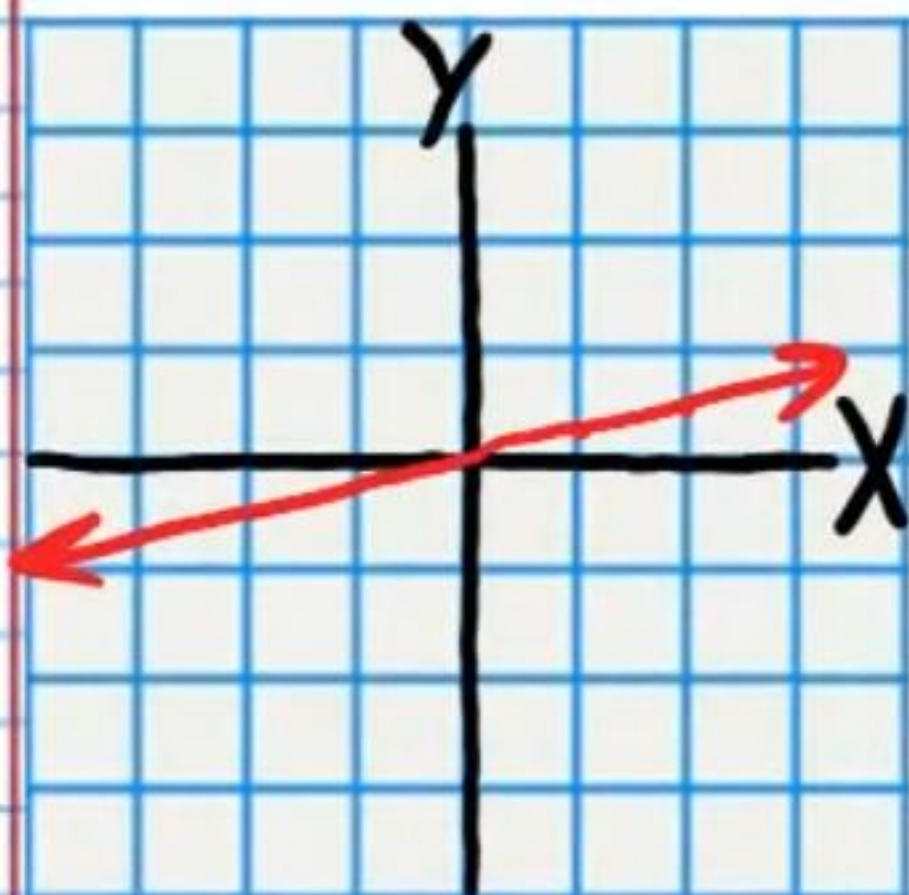
$$\tan(-45^\circ) = \frac{y}{x}$$

$$-1 = \frac{y}{x}$$

$$-x = y$$



⑮ $\theta = 10^\circ$



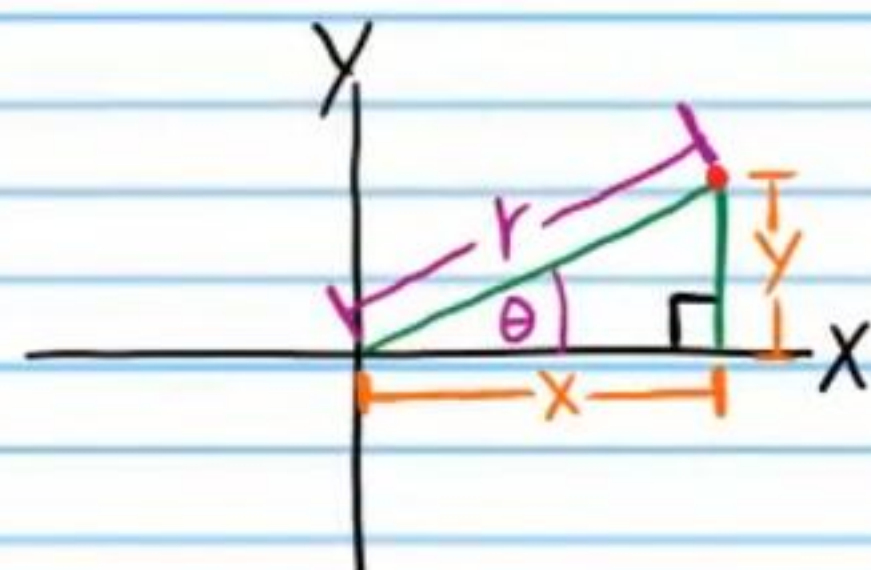
$$\tan \theta = \frac{y}{x}$$

$$\tan 10^\circ = \frac{y}{x}$$

$$0.176 \approx \frac{y}{x}$$

$$0.176x \approx y$$

Converting between Rectangular Equations and Polar Equations



$$x^2 + y^2 = r^2$$

$$\tan \theta = \frac{y}{x}$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

Convert to rectangular form.

• ⑰ $r = r^2$

$$r = x^2 + y^2$$

⑱ $r = \sin \theta + r \tan \theta$

$$r(r) = (\sin \theta + r \tan \theta) r$$

$$r^2 = r \sin \theta + r^2 \tan \theta$$

$$x^2 + y^2 = y + (x^2 + y^2) \frac{y}{x}$$

★ (21) $r = \cos\theta - 1$ Modifications will be necessary.

$$r(r) = (\cos\theta - 1)r$$

$$r^2 = r\cos\theta - r$$

$$x^2 + y^2 = x - \sqrt{x^2 + y^2}$$

$$\sqrt{x^2 + y^2} + x^2 + y^2 = x - \sqrt{x^2 + y^2} + \sqrt{x^2 + y^2}$$

$$\sqrt{x^2 + y^2} + x^2 + y^2 = x$$

$$\sqrt{x^2 + y^2} + x^2 + y^2 - x^2 - y^2 = x - x^2 - y^2$$

$$\sqrt{x^2 + y^2} = x - x^2 - y^2$$

$$(\sqrt{x^2 + y^2})^2 = (x - x^2 - y^2)^2$$

$$x^2 + y^2 = (x - x^2 - y^2)^2$$

• (22) $\tan^2\theta = r\cos\theta$

$$(\tan\theta)^2 = r\cos\theta$$

$$\left(\frac{y}{x}\right)^2 = x$$

$$\frac{y^2}{x^2} = x$$

$$(x^2) \frac{y^2}{x^2} = x(x^2)$$

$$y^2 = x^3$$

(23) $r = -3\sin\theta$

$$r(r) = (-3\sin\theta)r$$

$$r^2 = -3r\sin\theta$$

$$x^2 + y^2 = -3y$$

$$\textcircled{24} \quad r = 2 - \cos \theta$$

$$r(r) = (2 - \cos \theta)r$$

$$r^2 = 2r - r \cos \theta$$

$$x^2 + y^2 = 2\sqrt{x^2 + y^2} - x$$

$$\frac{x^2 + y^2 + x}{2} = \sqrt{x^2 + y^2}$$

$$\left(\frac{x^2 + y^2 + x}{2}\right)^2 = (\sqrt{x^2 + y^2})^2$$

$$\left(\frac{x^2 + y^2 + x}{2}\right)^2 = x^2 + y^2$$

Convert to polar form.

$$\bullet \textcircled{25} \quad 3xy = 1$$

$$3(r \cos \theta)(r \sin \theta) = 1$$

$$3r^2 \cos \theta \sin \theta = 1$$

$$3r^2 \frac{\sin(2\theta)}{2} = 1$$

$$\boxed{\frac{3}{2}r^2 \sin(2\theta) = 1}$$

Double Angle Formula

$$\sin(2A) = 2 \sin A \cos A$$

$$\frac{\sin(2A)}{2} = \frac{2 \sin A \cos A}{2}$$

$$\frac{\sin(2A)}{2} = \sin A \cos A$$

$$\textcircled{26} \quad x^2 + y^2 = x$$

$$r^2 = r \cos \theta$$

$$\frac{r^2}{r} = \frac{r \cos \theta}{r}$$

$$\boxed{r = \cos \theta}$$



$$(27) \frac{y}{x} = \sqrt{x^2 + y^2}$$

$$x^2 + y^2 = r^2$$

$$r = \pm \sqrt{x^2 + y^2}$$

θ will need restrictions.

$$\tan \theta = r, \quad 0 \leq \theta < 90^\circ \text{ or } 180^\circ \leq \theta < 270^\circ$$

$$(28) 5 + x = y$$

$$5 + r \cos \theta = r \sin \theta$$

$$5 = r \sin \theta - r \cos \theta$$

$$5 = r(\sin \theta - \cos \theta)$$

$$\frac{5}{\sin \theta - \cos \theta} = \frac{r(\sin \theta - \cos \theta)}{\sin \theta - \cos \theta}$$

$$r = \frac{5}{\sin \theta - \cos \theta}$$

$$(29) y = (x^2 + y^2)^2$$

$$r \sin \theta = (r^2)^2$$

$$r \sin \theta = r^4$$

$$\frac{r \sin \theta}{r} = \frac{r^4}{r}$$

$$\sin \theta = r^3$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$(30) \frac{x}{x^2 + y^2} = \left(\frac{y}{x}\right)^2 + 1$$

$$\frac{r \cos \theta}{r^2} = (\tan \theta)^2 + 1$$

$$\frac{\cos \theta}{r} = \tan^2 \theta + 1$$

$$\frac{\cos \theta}{r} = \sec^2 \theta$$

$$(v) \frac{\cos \theta}{r} = (\sec^2 \theta)(r)$$

$$\cos \theta = r \sec^2 \theta$$

$$\cos \theta = r \frac{1}{\cos^2 \theta}$$

$$(\cos^2 \theta) \cos \theta = \frac{r}{\cos^2 \theta} (\cos^2 \theta)$$

$$r = \cos^3 \theta$$